

A Practical Guide to Ensemble learning

TOPIC -- ENSEMBLE LEARNING

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Land cover mapping Land cover mapping is one of the major applications of Earth observation satellite sensors, using remote sensing and geospatial data, to identify the materials and objects which are located on the surface of target areas. Generally, the classes of target materials include roads, buildings, rivers, lakes, and vegetation. Some different ensemble learning approaches based on artificial neural networks, kernel principal component analysis (KPCA), decision trees with boosting, random forest and automatic design of multiple classifier systems, are proposed to efficiently identify land cover objects.

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$$y = \underset{c_j \in C}{\operatorname{argmax}} \sum_{h_i \in H} P(c_j | h_i) P(h_i | T) \quad \{\displaystyle y = \underset{c_j \in C}{\operatorname{argmax}} \sum_{h_i \in H} P(c_j | h_i) P(h_i | T)\}$$

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Face recognition, which recently has become one of the most popular research areas of pattern recognition, copes with identification or verification of a person by their digital images. Hierarchical ensembles based on Gabor Fisher classifier and independent component analysis preprocessing techniques are some of the earliest ensembles employed in this field.

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Ensemble size While the number of component classifiers of an ensemble has a great impact on the accuracy of prediction, there is a limited number of studies addressing this problem. A priori determining of ensemble size and the volume and velocity of big data streams make this even more crucial for online ensemble classifiers. Mostly statistical tests were used for determining the proper number of components. More recently, a theoretical framework suggested that there is an ideal number of component classifiers for an ensemble such that having more or less than this number of classifiers would deteriorate the accuracy. It is called "the law of diminishing returns in ensemble construction." Their theoretical framework shows that using the same number of independent component classifiers as class labels gives the highest accuracy.

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Cross-Validation Selection can be summed up as: "try them all with the training set, and pick the one that works best". Gating is a generalization of Cross-Validation Selection. It involves training another learning model to decide which of the models in the bucket is best-suited to solve the problem. Often, a perceptron is used for the gating model. It can be used to pick the "best" model, or it can be used to give a linear weight to the predictions from each model in the bucket. When a bucket of models is used with a large set of problems, it may be desirable to avoid training some of the models that take a long time to train. Landmark learning is a meta-learning approach that seeks to solve this problem. It involves training only the fast (but imprecise) algorithms in the bucket, and then using the performance of these algorithms to help determine which slow (but accurate) algorithm is most likely to do best.

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Distributed denial of service Distributed denial of service is one of the most threatening cyber-attacks that may happen to an internet service provider. By combining the output of single classifiers, ensemble classifiers reduce the total error of detecting and discriminating such attacks from legitimate flash crowds.

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The geometric framework Ensemble learning, including both regression and classification tasks, can be explained using a geometric framework. Within this framework, the output of each individual classifier or regressor for the entire dataset can be viewed as a point in a multi-dimensional space. Additionally, the target result is also represented as a point in this space, referred to as the "ideal point." The Euclidean distance is used as the metric to measure both the performance of a single classifier or regressor (the distance between its point and the ideal point) and the dissimilarity between two classifiers or regressors (the distance between their respective points). This perspective transforms ensemble learning into a deterministic problem. For example, within this geometric framework, it can be proved that the averaging of the outputs (scores) of all base classifiers or regressors can lead to equal or better results than the average of all the individual models. It can also be proved that if the optimal weighting scheme is used, then a weighted averaging approach can outperform

any of the individual classifiers or regressors that make up the ensemble or as good as the best performer at least.

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Overview Supervised learning algorithms search through a hypothesis space to find a suitable hypothesis that will make good predictions with a particular problem. Even if this space contains hypotheses that are very well-suited for a particular problem, it may be very difficult to find a good one. Ensembles combine multiple hypotheses to form one which should be theoretically better. Ensemble learning trains two or more machine learning algorithms on a specific classification or regression task. The algorithms within the ensemble model are generally referred as "base models", "base learners", or "weak learners" in literature. These base models can be constructed using a single modelling algorithm, or several different algorithms. The idea is to train a diverse set of weak models on the same modelling task, such that the outputs of each weak learner have poor predictive ability (i.e., high bias), and among all weak learners, the outcome and error values exhibit high variance. Fundamentally, an ensemble learning model trains at least two high-bias (weak) and high-variance (diverse) models to be combined into a better-performing model. The set of weak models — which would not produce satisfactory predictive results individually — are combined or averaged to produce a single, high performing, accurate, and low-variance model to fit the task as required. Ensemble learning typically refers to bagging (bootstrap aggregating), boosting or stacking/blending techniques to induce high variance among the base models. Bagging creates diversity by generating random samples from the training observations and fitting the same model to each different sample — also known as homogeneous parallel ensembles. Boosting follows an iterative process by sequentially training each base model on the up-weighted errors of the previous base model, producing an additive model to reduce the final model errors — also known as sequential ensemble learning. Stacking or blending consists of different base models, each trained independently (i.e. diverse/high variance) to be combined into the ensemble model — producing a heterogeneous parallel ensemble. Common applications of ensemble learning include random forests (an extension of bagging), Boosted Tree models, and Gradient Boosted Tree Models. Models in applications of stacking are generally more task-specific — such as combining clustering techniques with other parametric and/or non-parametric techniques. Evaluating the prediction of an ensemble typically requires more computation than evaluating the prediction of a single model. In one sense, ensemble learning may be thought of as a way to compensate for poor learning algorithms by performing a lot of extra computation. On the other hand, the alternative is to do a lot more learning with one non-ensemble model. An ensemble may be more

efficient at improving overall accuracy for the same increase in compute, storage, or communication resources by using that increase on two or more methods, than would have been improved by increasing resource use for a single method. Fast algorithms such as decision trees are commonly used in ensemble methods (e.g., random forests), although slower algorithms can benefit from ensemble techniques as well. By analogy, ensemble techniques have been used also in unsupervised learning scenarios, for example in consensus clustering or in anomaly detection.